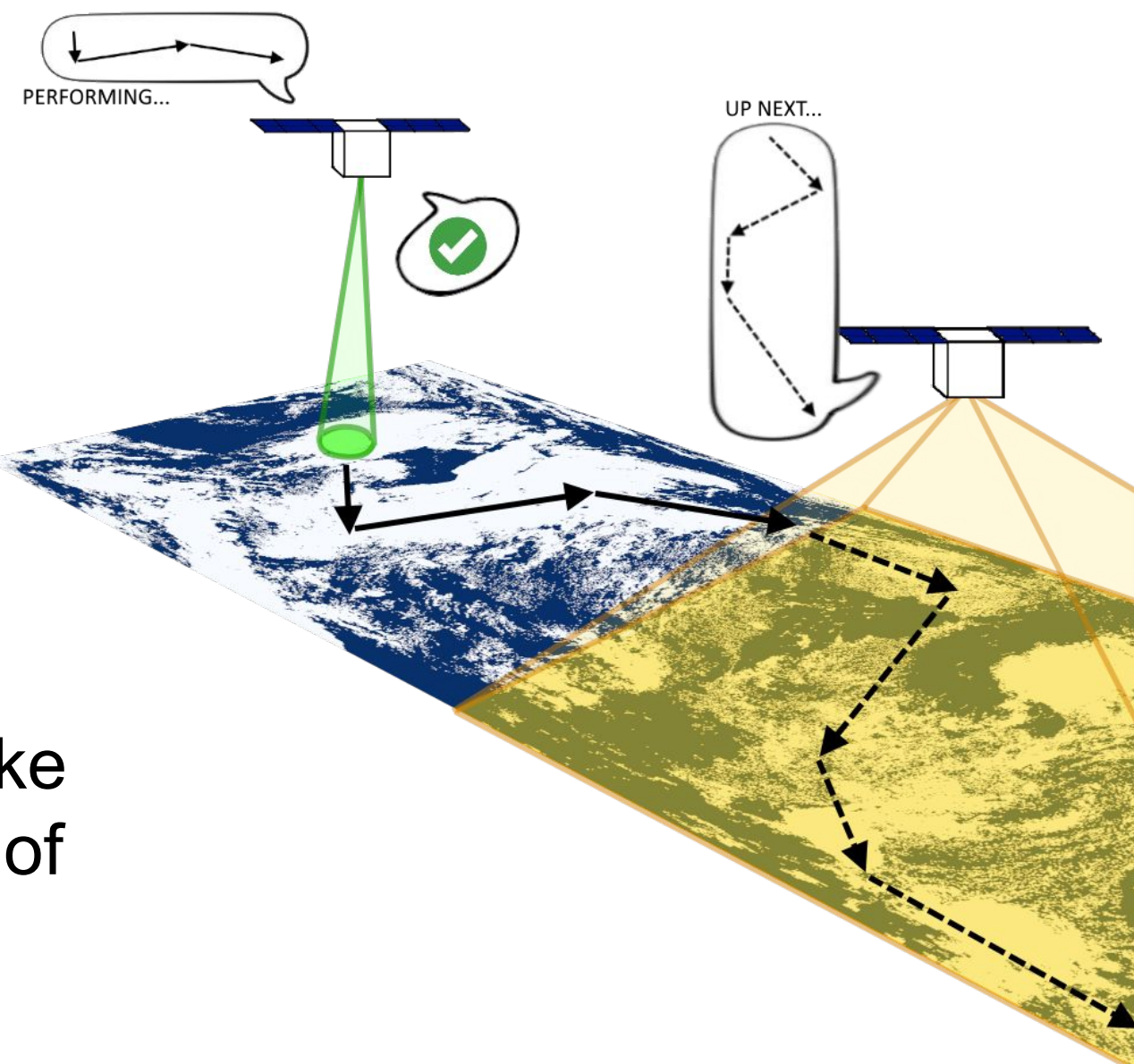




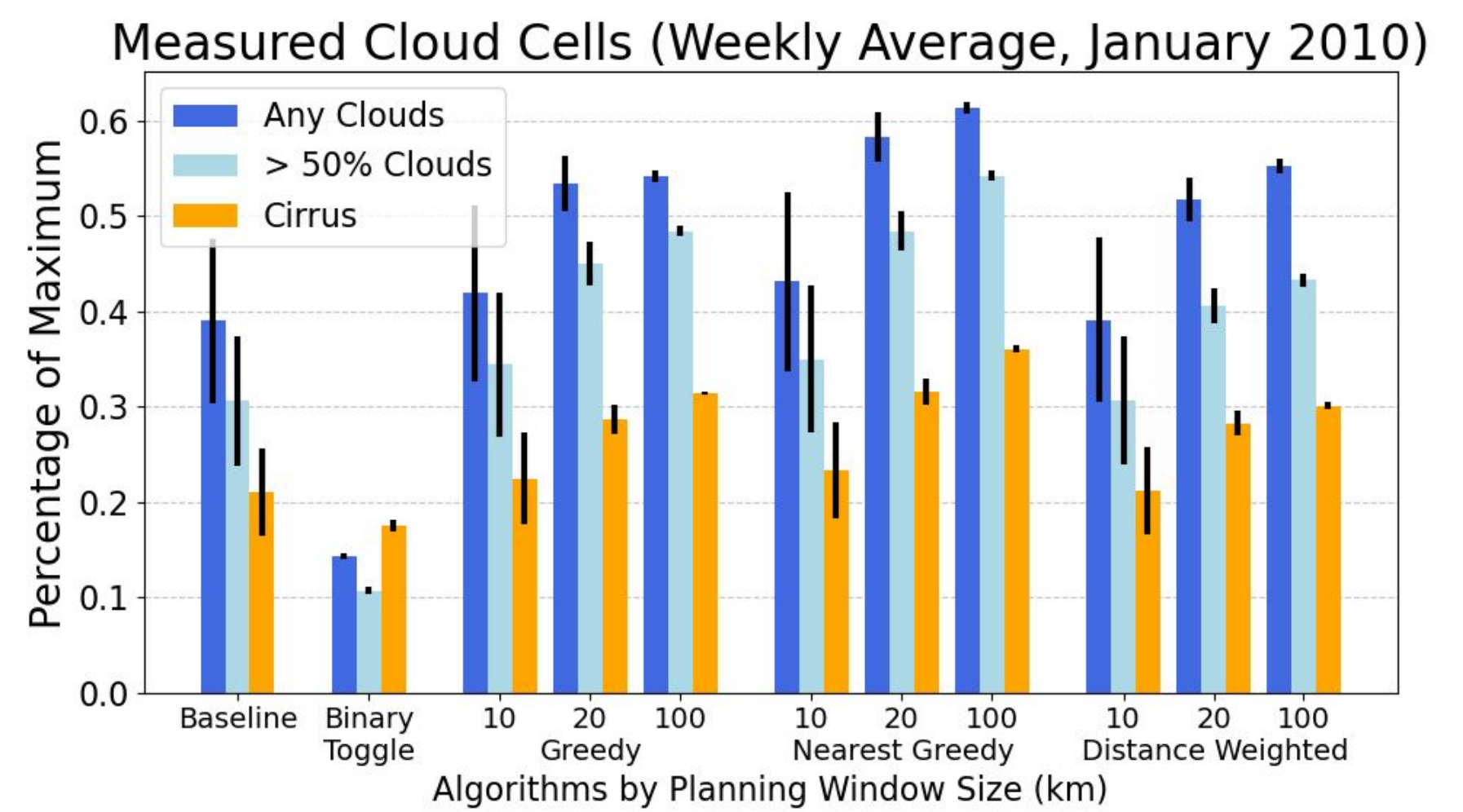
Introduction

Traditional cloud satellites waste **time** and **power** measuring empty sky, missing crucial, poorly understood clouds.

By using a **lookahead sensor**, a satellite can instead autonomously make **targeted measurements** of high value clouds [1].



Results

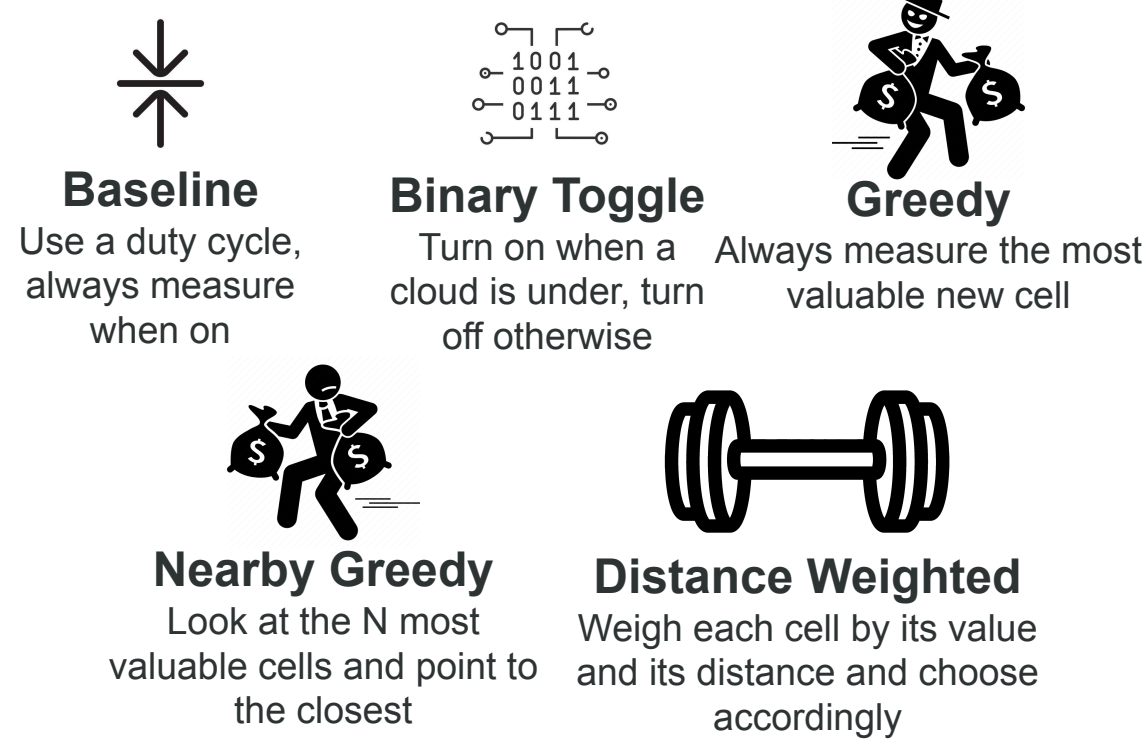
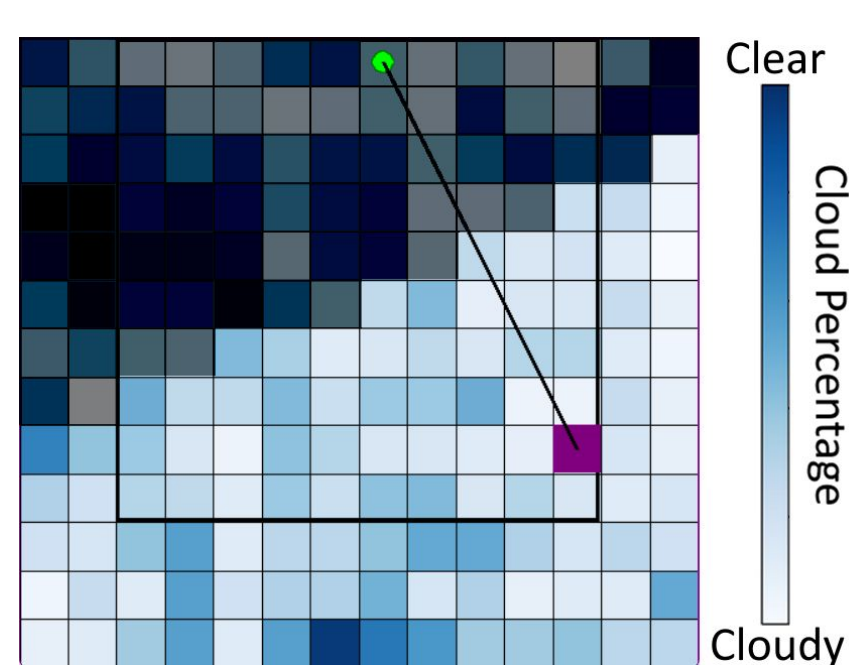


Algorithms with **20 km** or larger lookahead windows showed improvement over the baseline, both on average and for having less variance. **Nearest greedy** performed best.

Methods

Simulation for Algorithm Design

Using MODIS cloud masks, we built a simulation to test different prioritization algorithms for pointing at clouds.



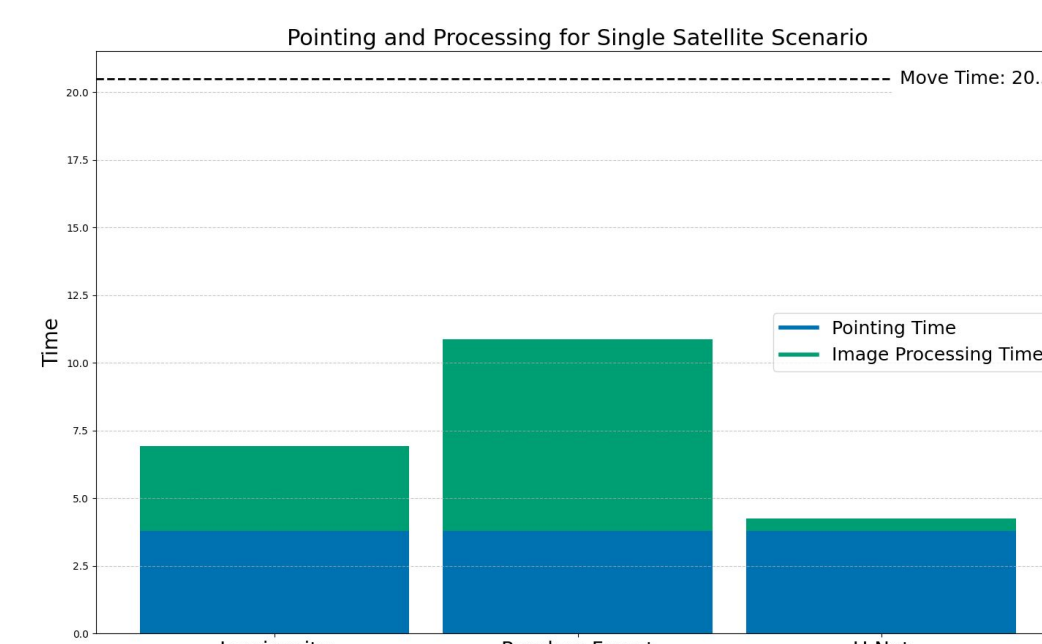
Lookahead Design

We analyzed strategies for taking lookaheads: a single satellite that points forward to image and scout satellites sending images to a main one.

Satellite Design

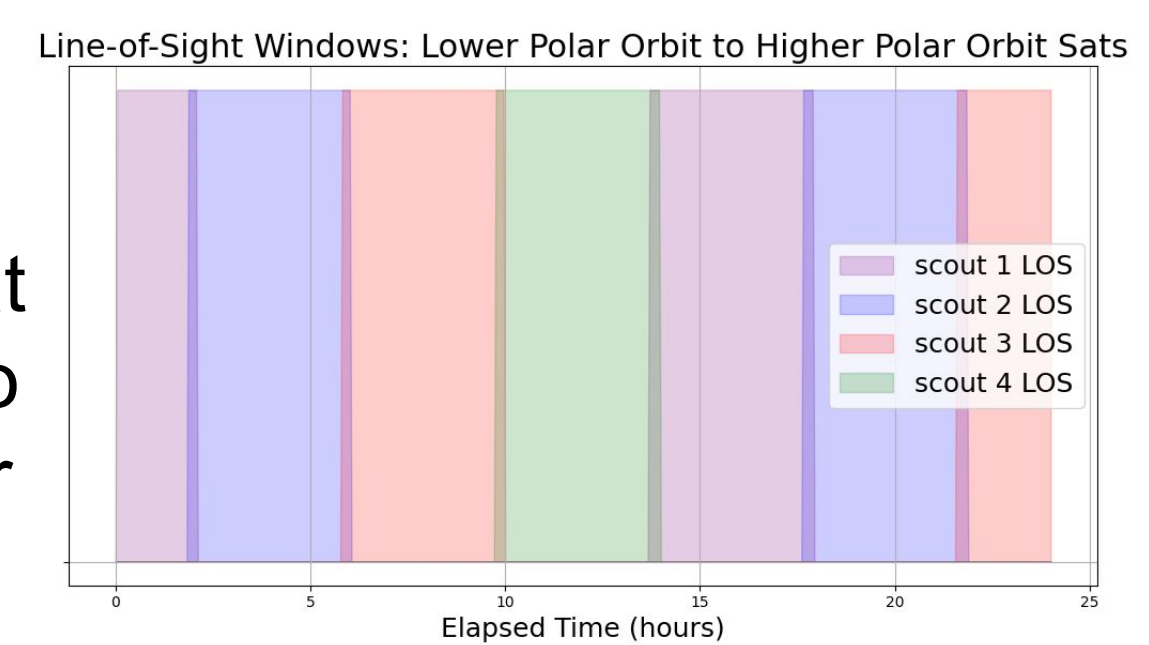
We identified high-value instruments for cloud science. We used a genetic algorithm to optimize “science per dollar.”

	CA	CTH	CTT	COD	AOD	LWP	r_e
Imager	1	0.6	0	0	0	0	0
Radar	0	0.6	0	0	0	0.4	0.4
Lidar	0	1	0	0.8	0.6	0.4	0.2
Radiometer	1	0.6	0.8	0.6	0	0.6	0.5
Polarimeter	0.2	0.8	0	0.4	0.8	0.4	0.4
MWR	0	0	0	0	0	0.8	0.2

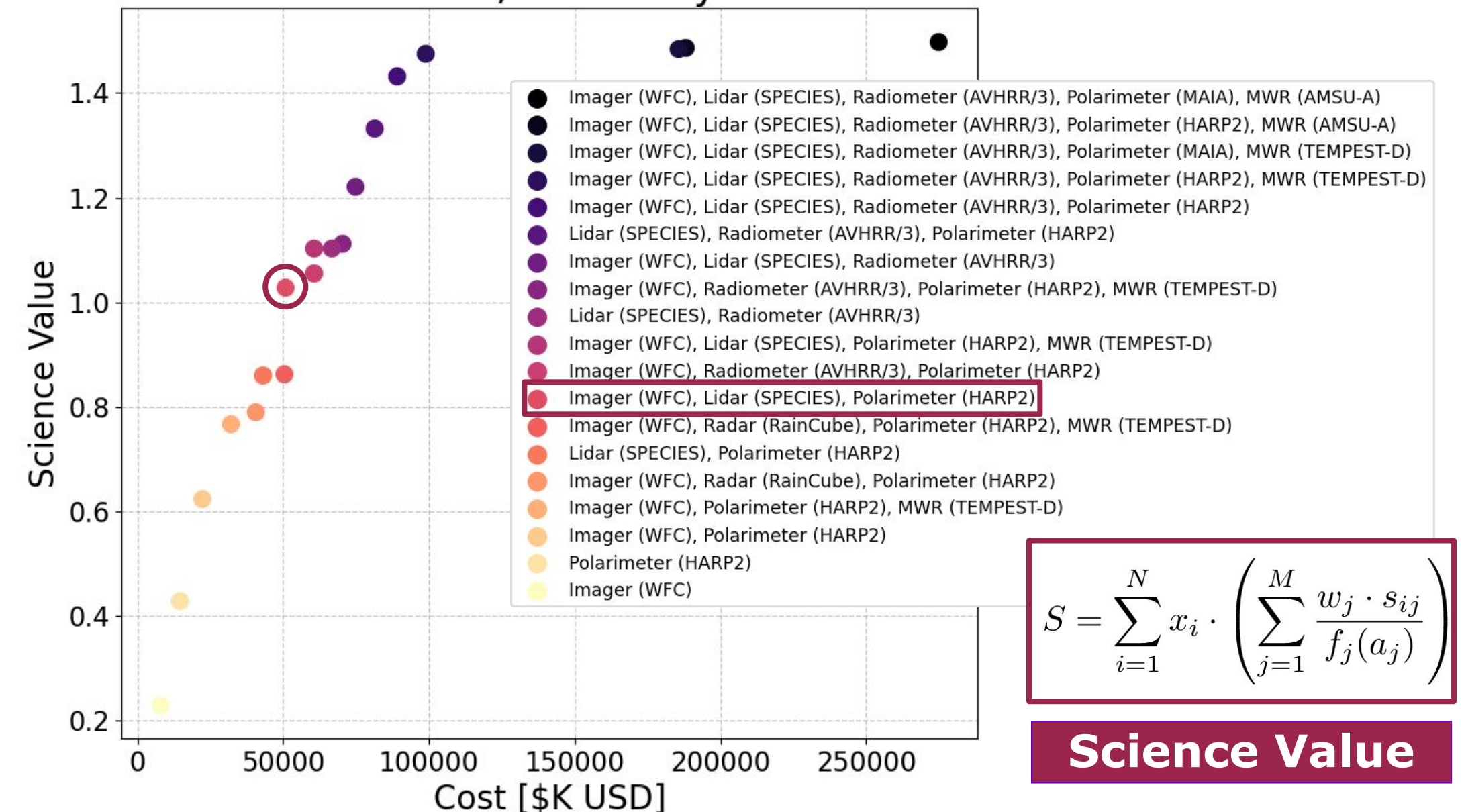


For a **single satellite** design, there is sufficient time to point forward and process a cloud image [2] before the satellite passes that area.

There are many configurations of scout satellites in **higher orbits** that offer consistent line of sight to the main satellite, allowing for larger **lookahead images**.



NSGA2 Pareto Front, Discovery Class Mission



$$S = \sum_{i=1}^N x_i \cdot \left(\sum_{j=1}^M \frac{w_j \cdot s_{ij}}{f_j(a_j)} \right)$$

Science Value

Polarimeters are frequently included in optimal designs.

Conclusions

Algorithm Performance

Prioritization algorithms need to balance **measuring high value cells** and **limiting pointing**.

Lookahead Configurations

Viable designs are **single satellite**, **one scout in the same orbit**, **four high-LEO scouts**, and **two GEO scouts**.

Satellite Instrumentation

The **imager, lidar, polarimeter** combination design best balances science value per dollar and will function in the dynamic framework.

Future Work

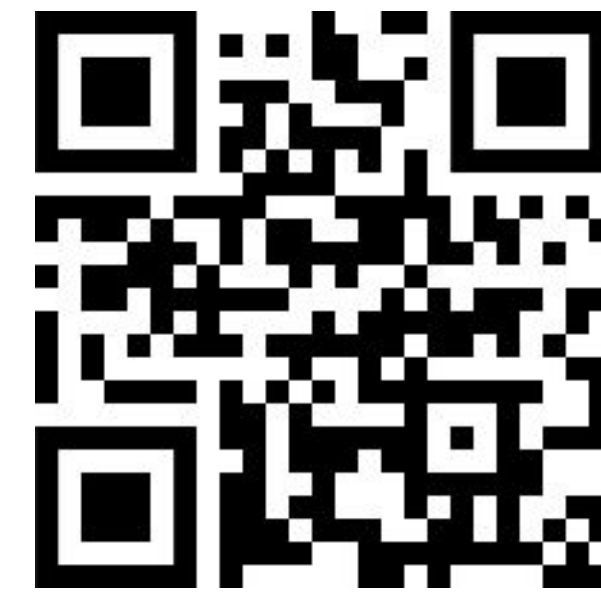
Simulation: Simulate the requirements for specific **instruments**, increase the number of **cloud targets**, implement **time to receive lookahead** to verify lookahead and satellite instrument choices.

Lookahead: Model **communications** and **image distortion**.

Satellite system design: Assess capabilities of instruments to measure **off-nadir** and penalize certain **combinations**, depending on capabilities.

References

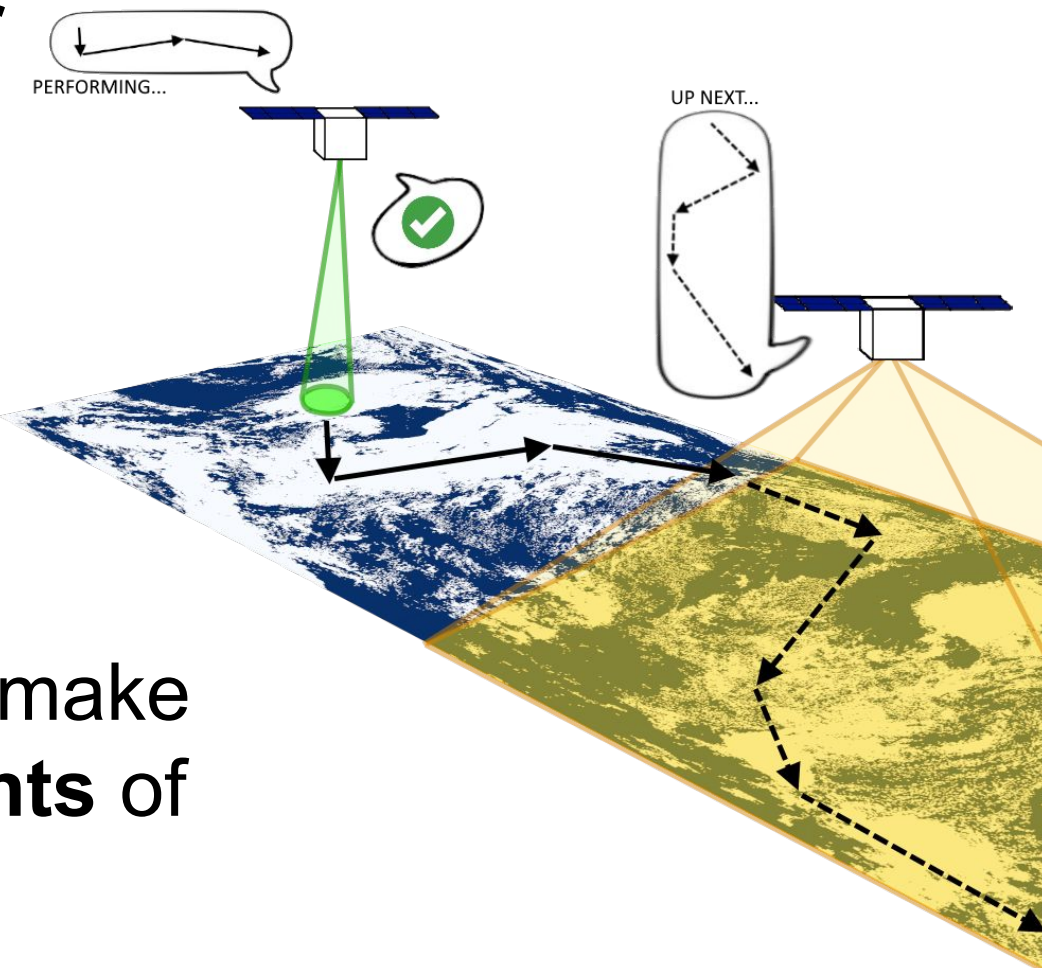
- [1] Dahl, Cahoy, “Improving cloud observations by autonomously pointing satellites,” *to appear in IEEE IGARSS*, Aug. 2025
- [2] Kacker, Meredith, Kusters, Tornio, Felt, Cahoy, “On-orbit rule-based and deep learning image segmentation strategies,” *AIAA SCITECH*, Jan. 2022.



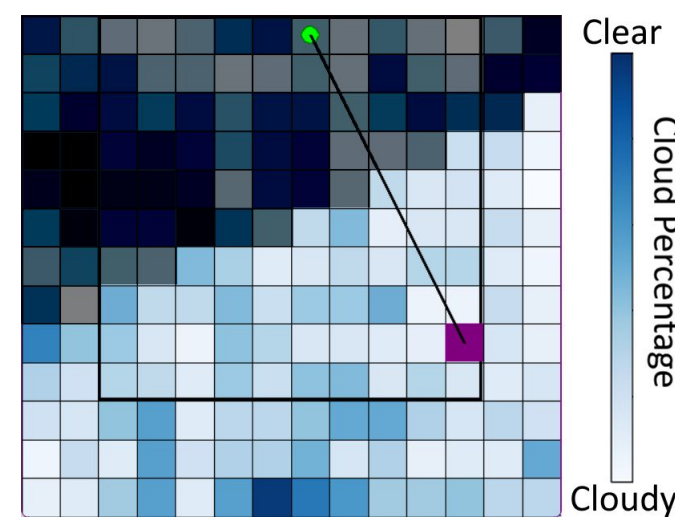
Introduction

Traditional cloud satellites waste **time** and **power** measuring empty sky, missing crucial, poorly understood clouds.

By using a **lookahead sensor**, a satellite can instead autonomously make **targeted measurements** of high value clouds [1].



Methods



Simulation for Algorithm Design

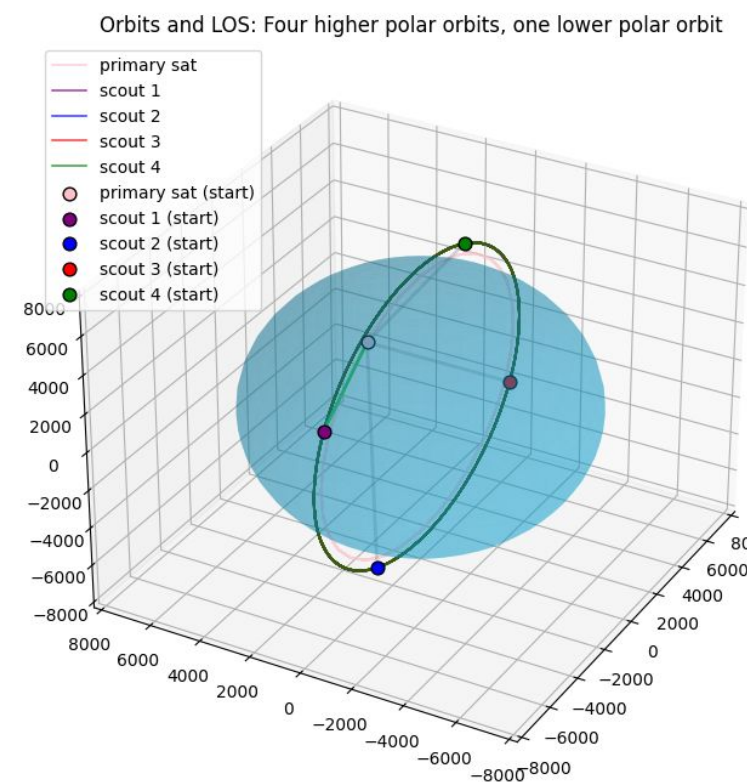
Using MODIS cloud masks, we built a simulation environment to test different prioritization algorithms for pointing at cloud cells.

Satellite Design

We identified high-value instruments for cloud science. We used a genetic algorithm to optimize “science per dollar.”

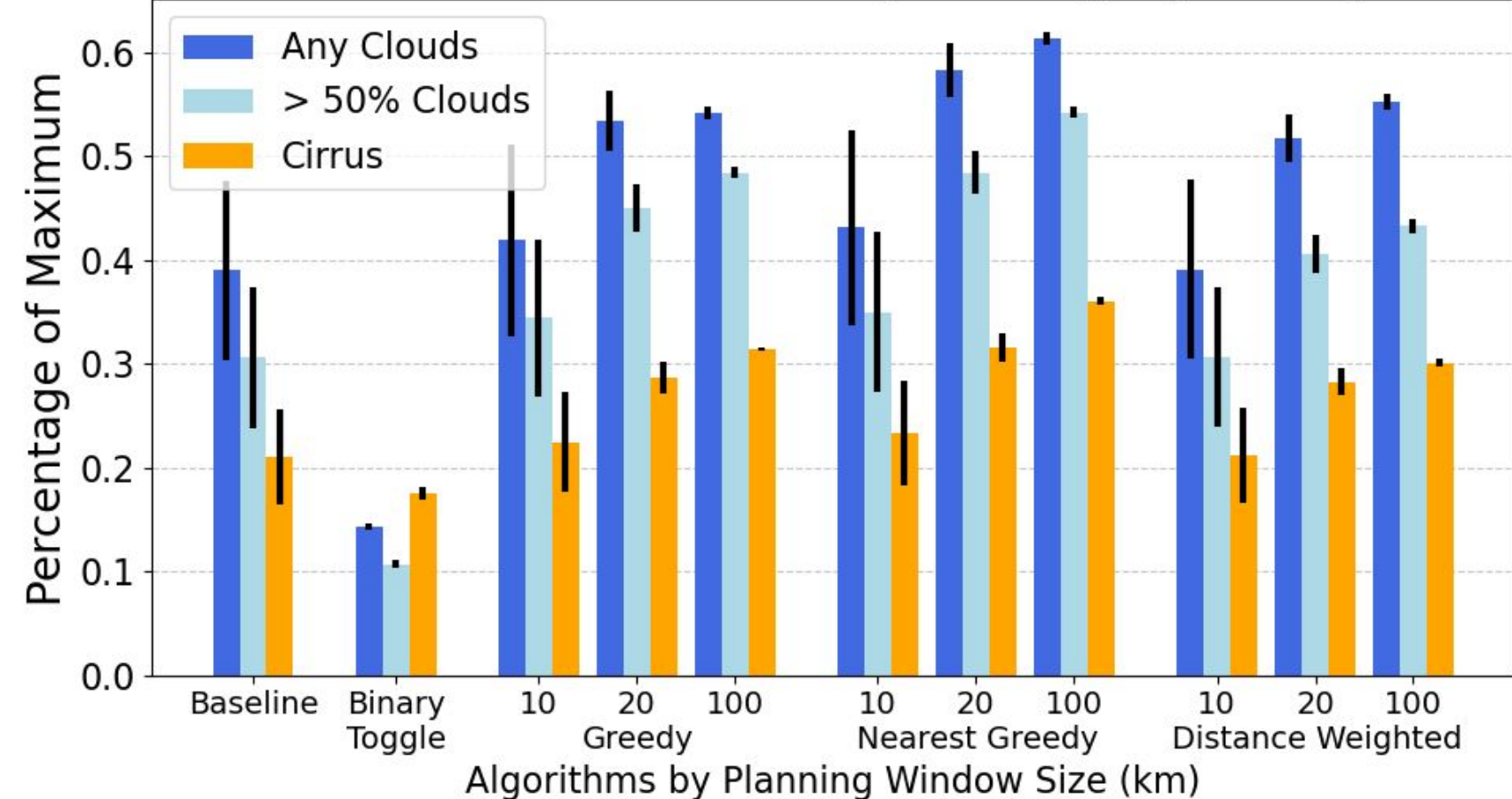
Lookahead Design

We analyzed strategies for making lookahead measurements: a single satellite or multiple satellites in different orbits.

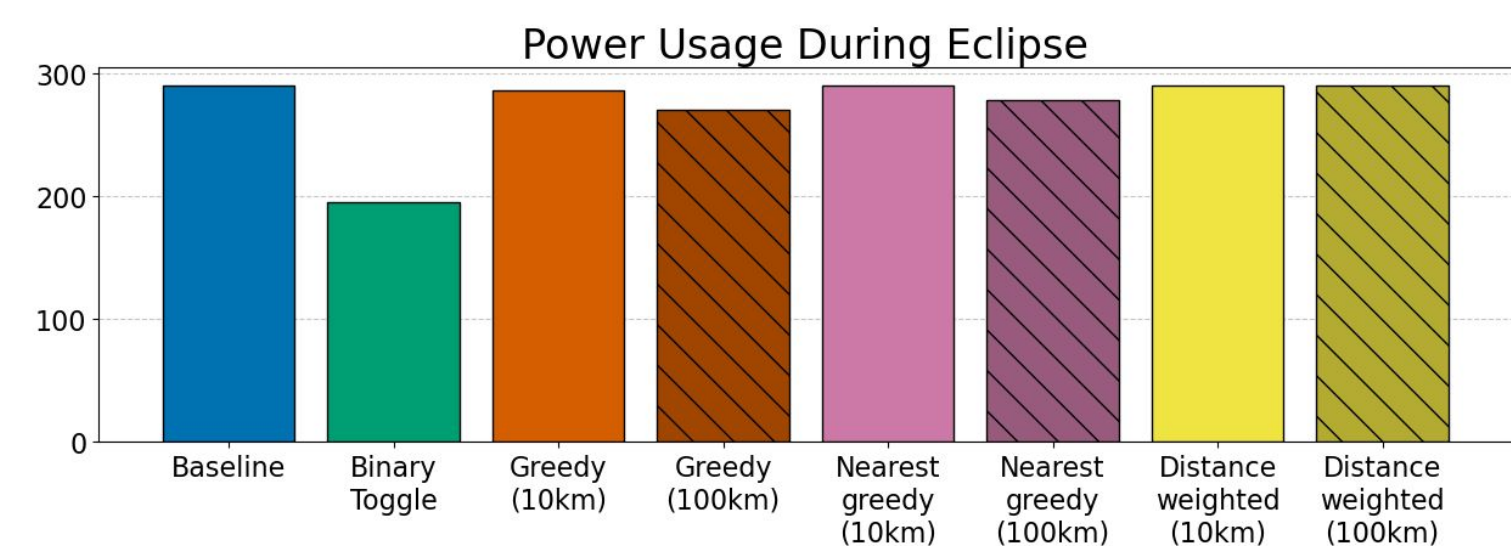
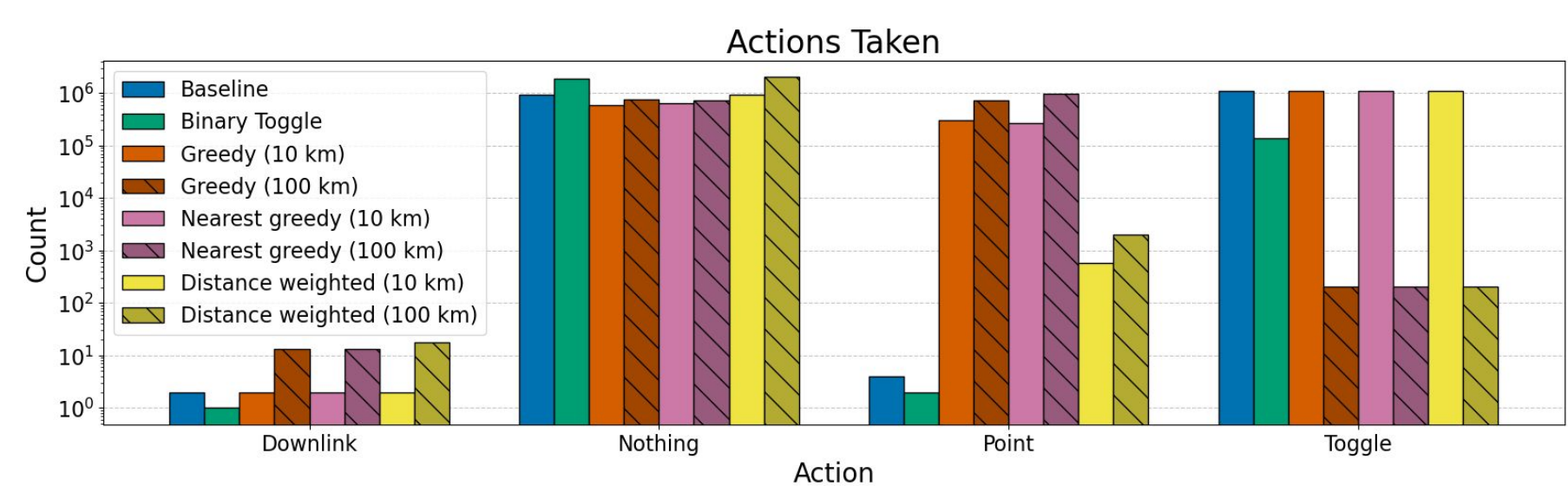


Results

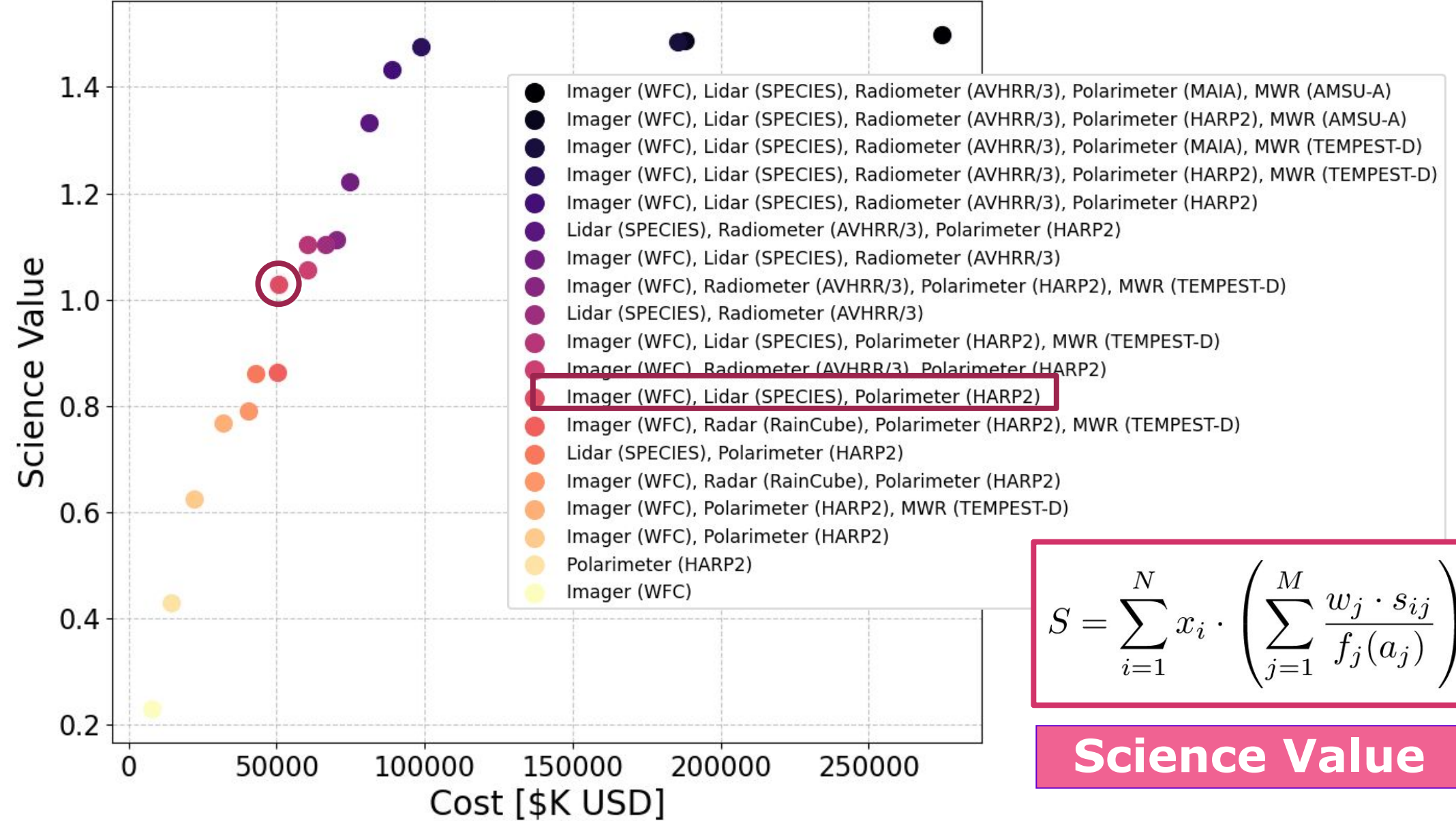
Measured Cloud Cells (Weekly Average, January 2010)



Algorithms with **20 km** or larger lookahead windows showed improvement over the baseline, both on average and for having less variance. **Nearest greedy** performed best.

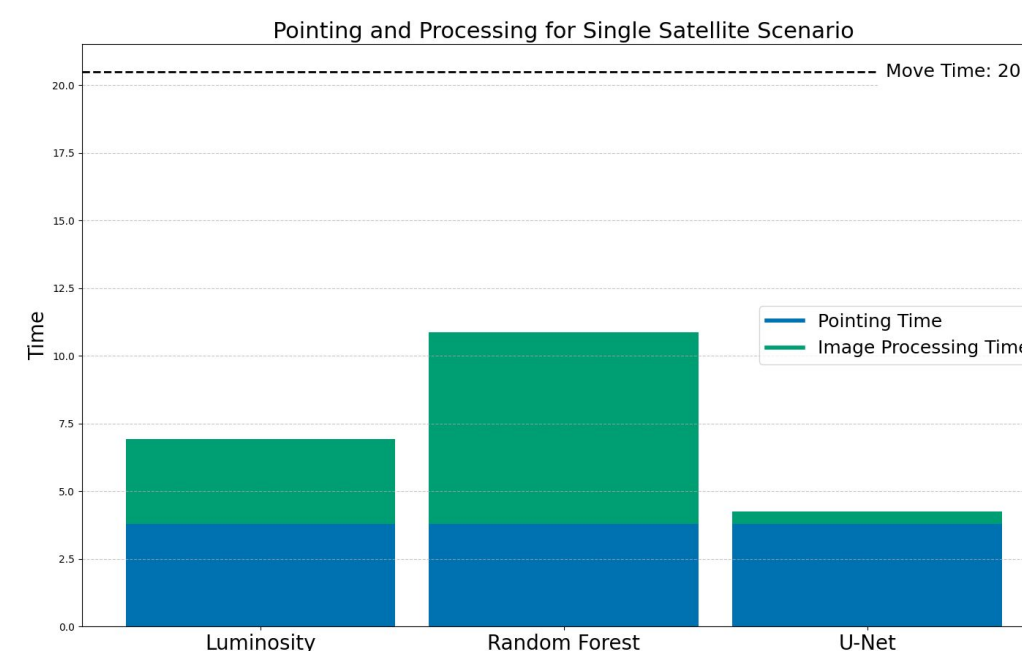


NSGA2 Pareto Front, Discovery Class Mission



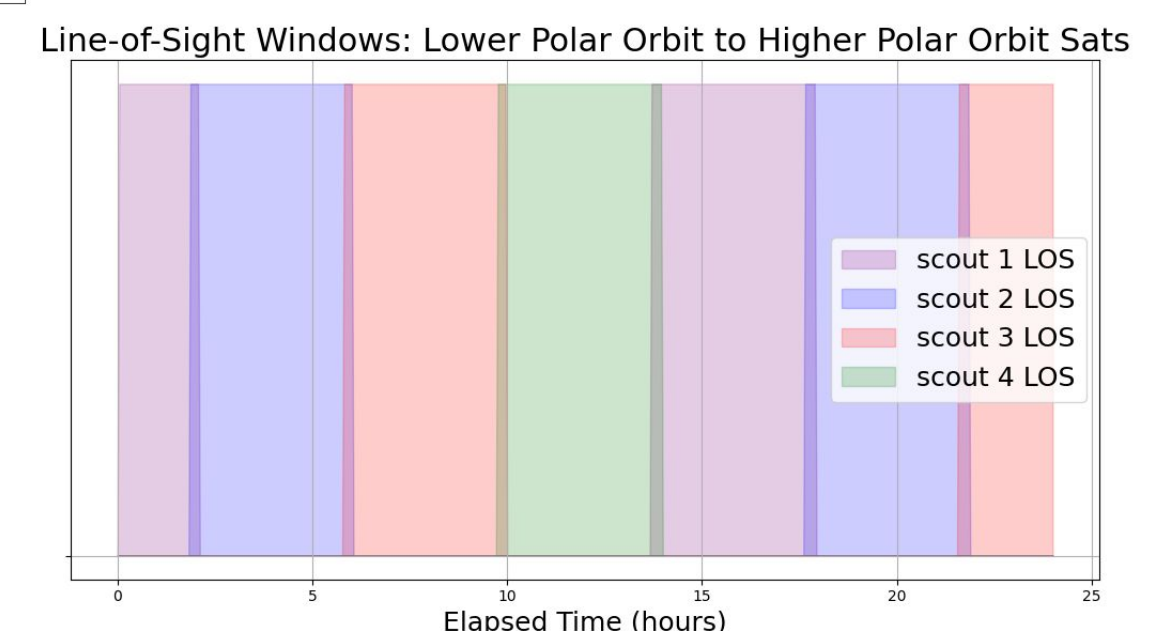
$$S = \sum_{i=1}^N x_i \cdot \left(\sum_{j=1}^M \frac{w_j \cdot s_{ij}}{f_j(a_j)} \right)$$

Science Value



For a **single satellite** design, there is sufficient time to point forward and process a cloud image [2] before the satellite passes that area.

Viable multiple satellite configurations include **four higher-LEO scouts** and **two GEO scouts**.



Conclusions

Algorithms

It is important for prioritization algorithms to both value **high value cells** and **limit the amount of pointing** to ensure the most time is spent measuring.

Satellite Configuration

We will be analyzing the **imager, lidar, polarimeter** combination design, as it balances science value per dollar and will function in the dynamic framework.

Lookahead Configuration

The candidate designs are **single satellite, four higher-LEO scouts, and two GEO scouts**.

Future Work

Simulation Updates

Updating the simulation to simulate specific instruments to confirm viability of potential mission architectures.

Model Updates

Relative capabilities of instruments to measure different properties

References

- [1] IGARSS paper
- [2] Shreeyam and Alex paper

